

Homework 1 Report

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Keywords: Principal Component Analyses, Linear Discriminant Analyses, Linear Discriminant Classifier, Quadratic Discriminant Classifier

Grade	Q1	Q2	Q3	Total
Max	4	3	3	10 pts
Expected	3	3	3	9 pts
Given				

The full source code for this homework, including all figures and scripts, can be found in the following GitHub repository:

https://github.com/drvshavva/BLG_527E_hw1

Q1) Dataset Generation and Classification

Generate a 2-dimensional, 2-class dataset with 1000 examples for each class where:

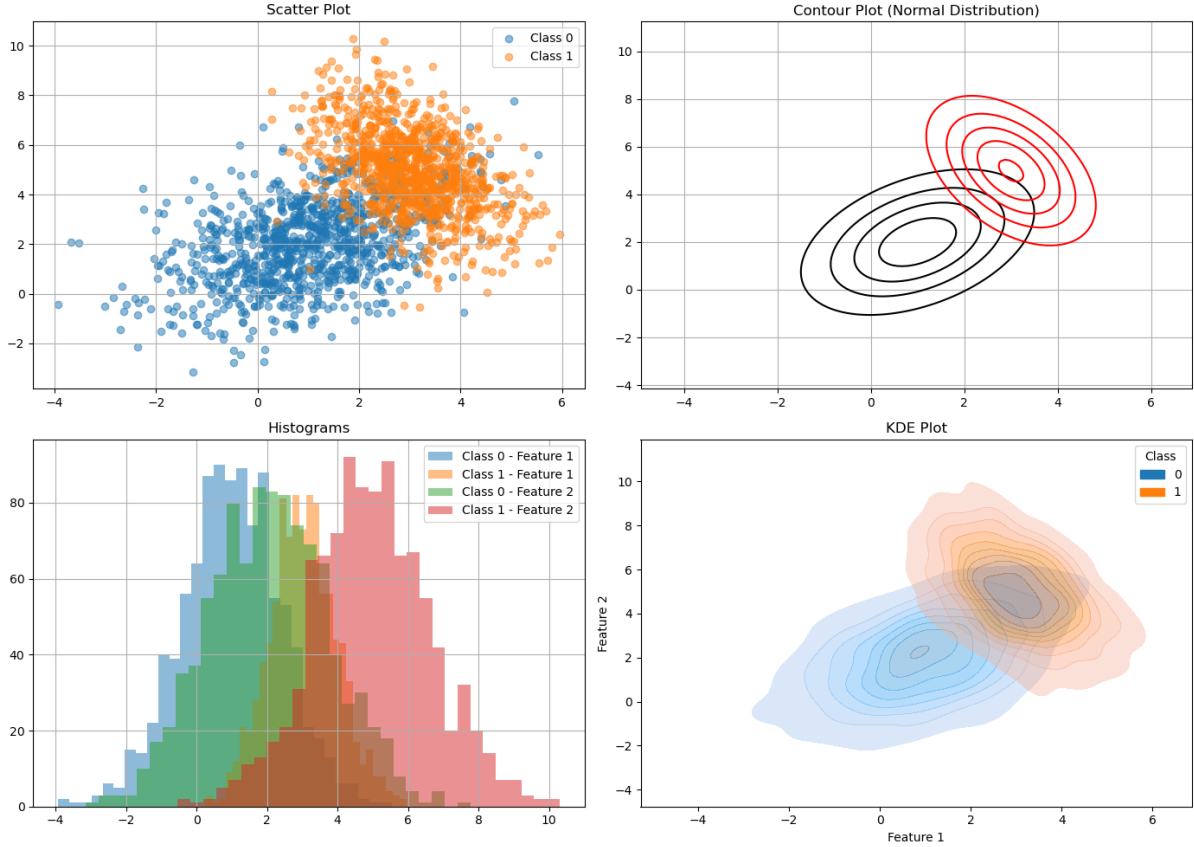
$$\mu_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mu_2 = \begin{bmatrix} 3 \\ 5 \end{bmatrix}, \quad \Sigma_1 = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \quad \Sigma_2 = \begin{bmatrix} 1 & -0.8 \\ -0.8 & 3 \end{bmatrix}$$

a) Plot the generated dataset:

The dataset is visualized using scatter plots, histograms, KDE plots, and contour plots. The two classes are clearly distinguishable in 2D space.

- The scatter plot illustrates the separability of the classes. With this plot, we can observe how widely the data is distributed and in which direction. Upon examining the figure, it is evident that the two different classes in our generated dataset are concentrated in distinct regions. There is a slight overlap in the central area, indicating that a classification model would not be able to achieve 100% accuracy.
- The contour plot displays the probability density contours for each class under the assumption of a normal distribution. It visualizes the covariance structure and orientation of the classes. When examining the plot, it is evident that the distributions of the classes are elliptical in different directions. In particular, Class 1 has tilted and tighter contours (indicating lower variance). This supports the preference for the quadratic discriminant classifier, as the classes have different covariance matrices.
- The histograms show the distribution of each feature for each class. They illustrate how discriminative each feature is for class separation. The more the histograms of the same feature differ between two classes, the more discriminative that feature is. In this case, both Feature 1 and Feature 2 can partially separate the classes. However, the histograms overlap to some extent, indicating that the separation is not perfect.

- The Kernel Density Estimation (KDE) plot provides a smooth estimation of class densities in the two-feature space. It highlights the regions with high data concentration more clearly. In our dataset, the kernel density centers of Class 0 and Class 1 are located in different regions. The contours do not significantly overlap, suggesting a good level of separability.



b) Training/Test Split and Estimation of Parameters:

For each class $c \in \{0, 1, \dots, K-1\}$, given a set of data samples $x_i \in \mathbb{R}^d$ such that $y_i = c$, the following computations are performed:

- Let n_c be the number of samples in class c . The mean vector $\mu_c \in \mathbb{R}^d$ is calculated as:

$$\mu_c = \frac{1}{n_c} \sum_{i=1}^{n_c} x_i$$

Each element $\mu_{c,j}$ of the mean vector is:

$$\mu_{c,j} = \frac{1}{n_c} \sum_{i=1}^{n_c} x_{i,j} \quad \text{for } j = 1, \dots, d$$

- The covariance matrix $\Sigma_c \in \mathbb{R}^{d \times d}$ for class c is calculated as:

$$\Sigma_c = \frac{1}{n_c - 1} \sum_{i=1}^{n_c} (x_i - \mu_c)(x_i - \mu_c)^T$$

Each element $\Sigma_{c,jk}$ of the matrix is:

$$\Sigma_{c,jk} = \frac{1}{n_c - 1} \sum_{i=1}^{n_c} (x_{i,j} - \mu_{c,j})(x_{i,k} - \mu_{c,k})$$

The dataset is split into 80% training and 20% testing data.

From the training data, the mean vectors and covariance matrices for each class are calculated as follows:

$$\mu_0 = \begin{bmatrix} 0.9878 \\ 2.0858 \end{bmatrix}, \quad \Sigma_0 = \begin{bmatrix} 2.0559 & 1.0646 \\ 1.0646 & 3.0051 \end{bmatrix}$$

$$\mu_1 = \begin{bmatrix} 3.0225 \\ 4.9585 \end{bmatrix}, \quad \Sigma_1 = \begin{bmatrix} 0.8889 & -0.7653 \\ -0.7653 & 2.9454 \end{bmatrix}$$

c) Quadratic Discriminant Classifier (QDC): For each class k , the discriminant function is defined as:

$$g_k(\mathbf{x}) = -\frac{1}{2} \log |\Sigma_k| - \frac{1}{2} (\mathbf{x} - \mu_k)^T \Sigma_k^{-1} (\mathbf{x} - \mu_k) + \log P(\omega_k)$$

Term	Meaning
μ_k	Mean vector of class k
Σ_k	Covariance matrix of class k
$P(\omega_k)$	Prior probability of class k
Σ_k^{-1}	Inverse of the covariance matrix
$(\mathbf{x} - \mu_k)^T \Sigma_k^{-1} (\mathbf{x} - \mu_k)$	Squared Mahalanobis distance

An observation $\mathbf{x}$ is assigned to the class with the maximum discriminant value:

$$\hat{y} = \arg \max_k g_k(\mathbf{x})$$

For two-class classification:

$$g_1(\mathbf{x}) = g_2(\mathbf{x}) \Rightarrow \text{decision boundary}$$

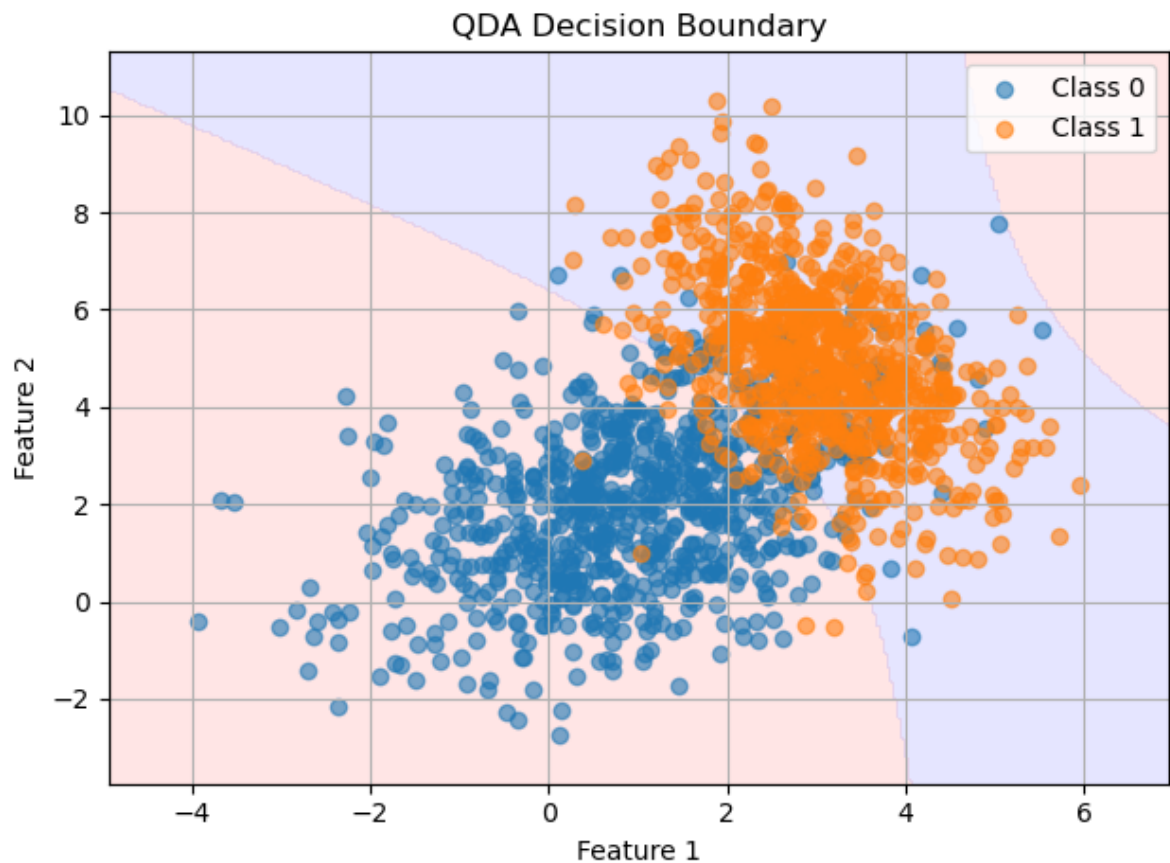
This results in a **quadratic** decision boundary, since each class may have a different covariance matrix. If $\Sigma_k = \Sigma$ for all k , the model reduces to the LDA (Linear Discriminant Analysis), and the decision boundary becomes linear.

Classification error is calculated as the fraction of incorrect predictions:

$$\text{Error} = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(y_i \neq \hat{y}_i)$$

Using class-specific covariance matrices, a quadratic discriminant classifier is constructed.

- Train Error (QDA): 0.1075
- Test Error (QDA): 0.0825



d) Linear Discriminant Classifier (LDC): Linear Discriminant Analysis assumes that:

- Each class shares a **common covariance matrix**:

$$\Sigma_k = \Sigma \quad \forall k$$

- Each class has its own mean vector μ_k .

Under these assumptions, the resulting decision boundaries are **linear**.

For each class k , the discriminant function is given by:

$$g_k(\mathbf{x}) = \mathbf{x}^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log P(\omega_k)$$

Term	Description
μ_k	Mean vector of class k
Σ	Shared covariance matrix (for all classes)
$P(\omega_k)$	Prior probability of class k
$\mathbf{x}$	Observation (feature vector)

The shared covariance matrix Σ is computed as:

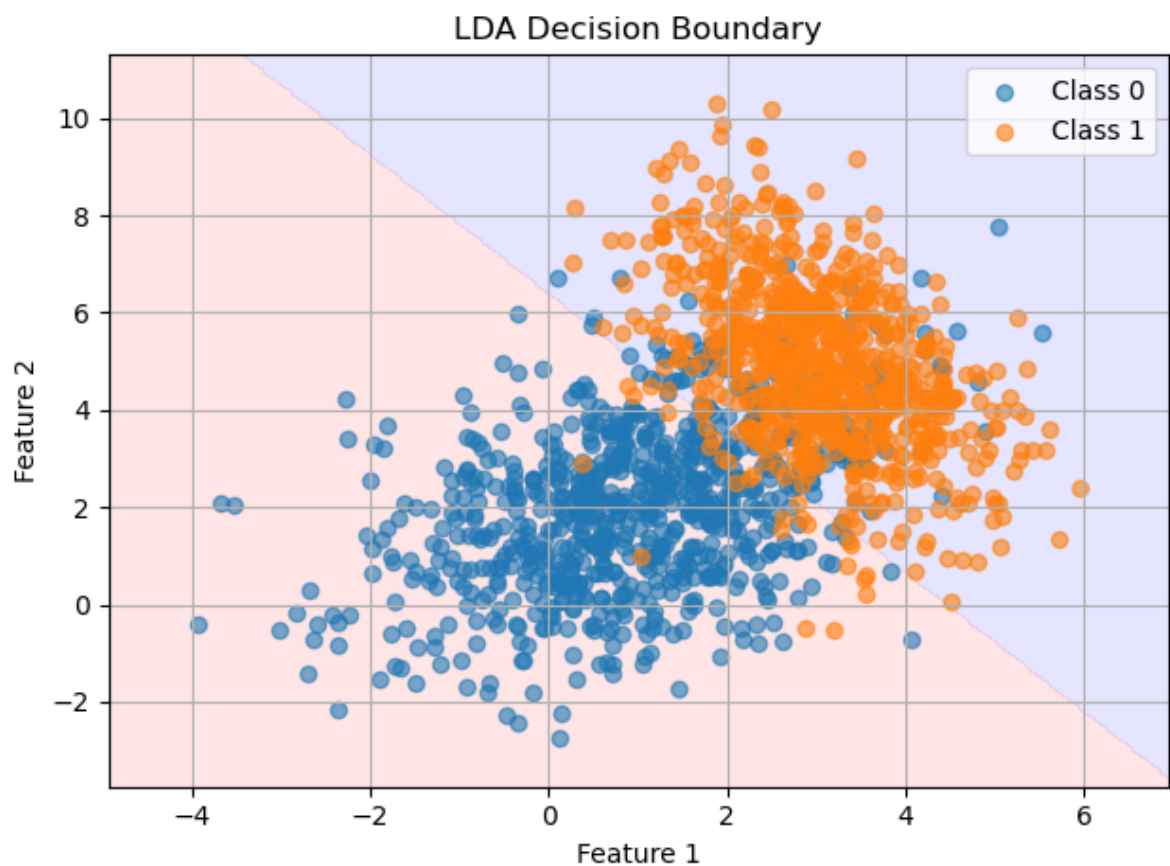
$$\Sigma = \frac{1}{N - K} \sum_{k=1}^K \sum_{i=1}^{N_k} \left(x_i^{(k)} - \mu_k \right) \left(x_i^{(k)} - \mu_k \right)^T$$

Assume:

- K : total number of classes (e.g., 2 classes: 0 and 1)
- X_k : data points from class k
- μ_k : mean vector of class k
- N_k : number of samples in class k
- $N = \sum_{k=1}^K N_k$: total number of samples

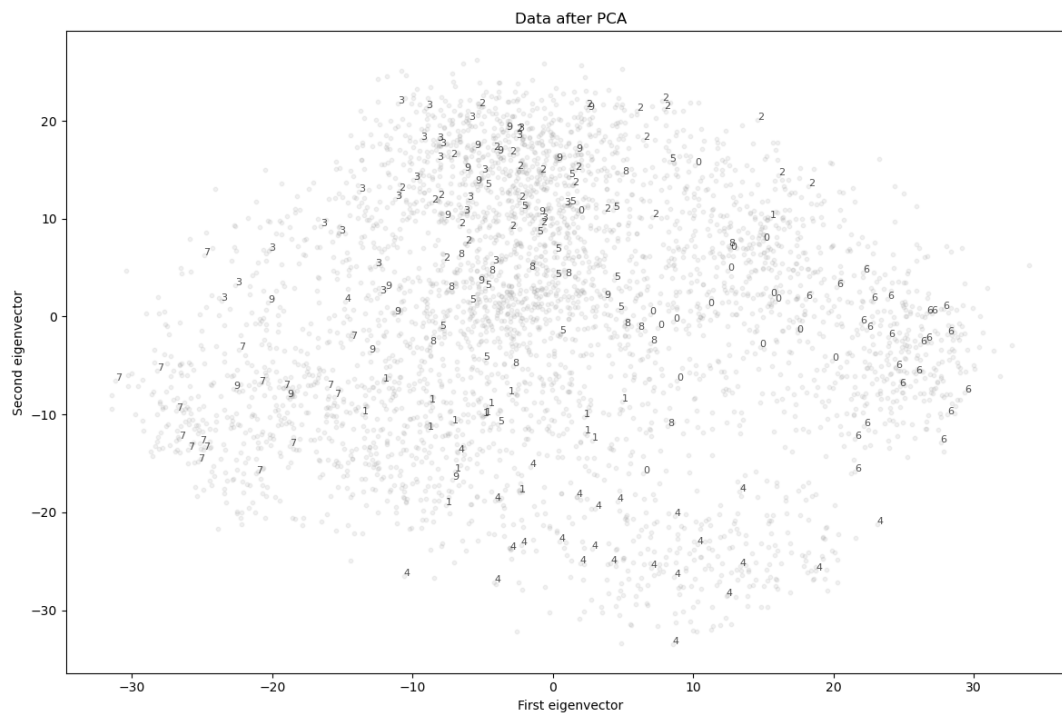
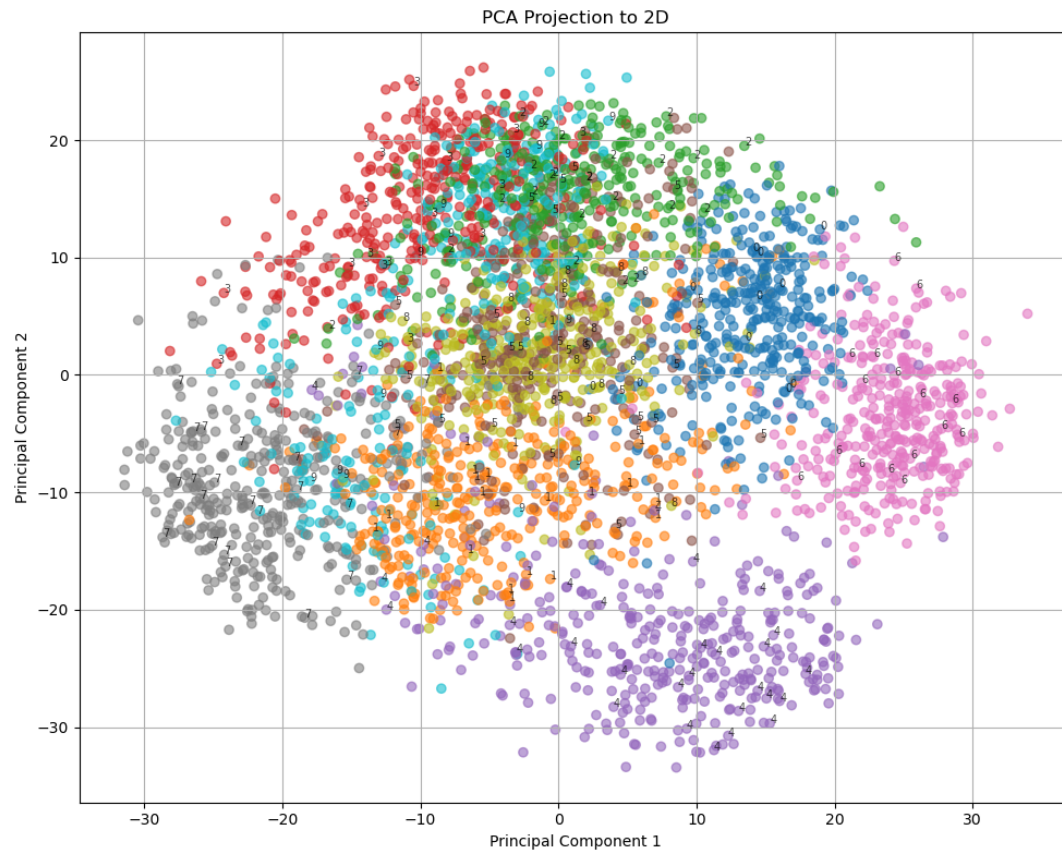
Using a shared covariance matrix computed as the weighted average of class covariances, a linear discriminant classifier is constructed.

- Train Error (LDA): 0.1087
- Test Error (LDA): 0.0900



Q2) Digits Dataset Principal Component Analysis (PCA)

In this section, we implement Principal Component Analysis (PCA) from scratch, using the `opdigits` dataset provided in `data.txt`. The goal of PCA is to reduce the dimensionality of the data while preserving as much variance as possible by projecting the data onto a new set of orthogonal axes (principal components).



1. **Compute the Mean Vector Manually:**

$$\mu_j = \frac{1}{n} \sum_{i=1}^n x_{ij}, \quad \text{for } j = 1, \dots, d$$

2. **Center the Data** (mean normalization):

$$\tilde{x}_{ij} = x_{ij} - \mu_j$$

3. Compute the Covariance Matrix Manually:

$$\Sigma_{jk} = \frac{1}{n-1} \sum_{i=1}^n \tilde{x}_{ij} \tilde{x}_{ik}$$

4. Compute Eigenvalues and Eigenvectors:

$$\Sigma \mathbf{v}_j = \lambda_j \mathbf{v}_j$$

Sort the eigenvalues in descending order and select the top 2 eigenvectors.

5. Project Data onto Principal Components:

$$Z = \tilde{X} \cdot W$$

where $W \in \mathbb{R}^{d \times 2}$ is the matrix containing the two principal eigenvectors.

Q3) Digits Dataset Linear Discriminant Analysis (LDA)

Linear Discriminant Analysis projects high-dimensional data into a lower-dimensional space (here, 2D) that maximizes class separability using label information. Unlike PCA, which captures variance, LDA focuses on between-class vs. within-class scatter. The resulting 2D projection shows better clustering by class. Annotating 200 digits helps visually verify label separation.

LDA seeks a linear projection that maximizes class separability by computing two matrices:

- **Within-class scatter matrix:**

$$S_W = \sum_{k=1}^K \sum_{x_i \in C_k} (x_i - \mu_k)(x_i - \mu_k)^T$$

- **Between-class scatter matrix:**

$$S_B = \sum_{k=1}^K N_k (\mu_k - \mu)(\mu_k - \mu)^T$$

The optimal projection matrix W is obtained by solving:

$$S_W^{-1} S_B \vec{w} = \lambda \vec{w}$$

and selecting the eigenvectors corresponding to the top eigenvalues.

